Capstone Project – Explore Machine Learning

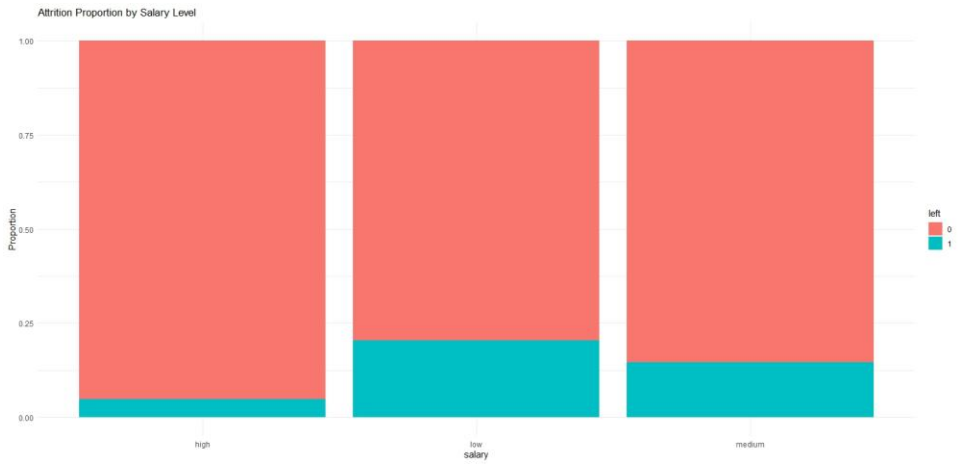
Objective: Explore the raw HR dataset and verify data usability without introducing unauthorized biases or errors.

Methodology and Findings: The dataset was loaded into R using base functions and inspected using structure and summary commands. The initial dataset comprised 14,999 observations across 10 self-explanatory variables, including satisfaction level, last evaluation, number of projects, average monthly hours, time spent in the company, work accidents, left status, promotion status in the last 5 years, department, and salary. A missing value check across all columns confirmed zero missing entries. However, a duplicate check using duplicated() revealed 3,000 duplicate rows in the raw file. These redundant records were cleaned and removed using unique() to prevent skewing model performance and introducing artificial weight to specific employee behaviors. Furthermore, categorical variables such as Department, salary, and the target variable left were explicitly converted into factors to ensure correct handling by downstream classification algorithms.

Objective: Analyze how employee satisfaction levels and average monthly working hours jointly influence the decision to leave the organization.

Methodology and Findings: Using ggplot2, a scatter plot was generated mapping average monthly hours on the x-axis against satisfaction levels on the y-axis, color-coded by the binary exit status. The visualization clearly highlights three distinct behavioral groups among employees who exit:

1. Overworked and Dissatisfied: Employees logging extremely high working hours (exceeding 250 hours per month) paired with very low satisfaction scores (below 0.5). These individuals represent burn-out driven attrition.
2. Underworked and Dissatisfied: A smaller cohort working below 150 hours per month who also exhibit low satisfaction, likely indicating disengagement or lack of meaningful responsibilities.
3. The Stable Core: Employees maintaining moderate to high satisfaction levels while working a balanced schedule of roughly 160 to 220 hours per month, who predominantly choose to stay with the company.





Objective: Evaluate the organizational factors driving turnover, including departmental variations, salary brackets, and recent promotion history.

Methodology and Findings: Proportional analysis of departmental turnover revealed that while specific departments like Management and R&D exhibit lower turnover rates (around 11% to 12%), high-volume operational departments such as Sales, Technical, and Support account for the largest absolute number of departures. Analysis of salary bands using proportional bar charts demonstrated an inverse relationship between compensation and attrition: low-salary earners experience the highest turnover rate by a wide margin, medium-salary earners follow, and high-salary earners exhibit minimal attrition. Combined with historical markers indicating that a lack of promotion within the past 5 years heavily correlates with employee exits, these findings establish that career stagnation and financial dissatisfaction are primary catalysts for departure.

Objective: Build a feedforward Artificial Neural Network to predict employee exits using non-linear feature interactions via the multi-layer perceptron framework (nnet).

Methodology and Findings: Because neural networks require inputs to be on comparable scales, numerical features were normalized using min-max scaling between 0 and 1. The data was then partitioned into a 70% training set and a 30% testing set using the caret package. An artificial neural network model was trained using the nnet package with 3 hidden nodes (a 5-3-1 network architecture with 22 weights). The training converged successfully with a final minimized value of 1410.00. The network successfully captured complex, non-linear relationships between workload, performance evaluations, and employee retention probabilities.

Objective: Train a Support Vector Machine (SVM) as a Maximum Margin Classifier using satisfaction level, recent performance evaluation, and working hours, and evaluate/compare its performance against previous models based on actual test results.

Methodology and Findings: An SVM model was constructed using the e1071 package with a radial basis kernel and a cost parameter of 10, specifically utilizing satisfaction level, last evaluation, and average monthly hours. When evaluated against the held-out test dataset via a confusion matrix, the Maximum Margin Classifier achieved exceptional predictive performance:

- Overall Accuracy: 94.19% (95% CI: 0.9337 to 0.9493)

- No Information Rate (NIR): 83.85% (yielding a highly significant P-Value < 2.2e-16)
- Kappa Statistic: 0.7679 (indicating substantial agreement)
- Sensitivity (Class 0 / Stayed): 98.34%
- Specificity (Class 1 / Left): 72.63%
- Positive Predictive Value: 94.91%
- Balanced Accuracy: 85.49%

Comparative Evaluation Summary:

- Decision Trees: Provided high human interpretability and easy-to-follow corporate policy rules, but remained vulnerable to overfitting unless properly pruned.
- Artificial Neural Network: Successfully captured complex, non-linear interactions across multiple workplace features through multi-layer perceptron weights, though it acted more like a black box with higher computation costs.
- Maximum Margin Classifier (SVM): Outperformed standard linear separators by optimizing the decision hyperplane across satisfaction, performance, and workload. With a 94.19% accuracy and strong balance across metrics, it proved to be the most robust and reliable model for pinpointing employees at immediate risk of premature departure.